THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE · HARD LEVEL FULL-YEAR PAPER SET 3

Full Syllabus (Chapters 1–13) | RBSE Pattern | Session 2026–27

Time: 3 Hours Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–26 are Very Short Answer Type (2 marks each).
4. Q. 27–32 are Short Answer Type (3 marks each).
5. Q. 33–38 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 31–32 and Q. 36–38.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. Assertion (A): Chloroplasts are found in plant cells but not in animal cells. Reason (R): Plants make their own food while animals depend on others. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

2. Assertion (A): Vaccination protects a child against a disease before exposure. Reason (R): Vaccines make the body produce antibodies against the pathogen. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

3. Assertion (A): A fuse is connected in series with the appliance. Reason (R): A fuse must carry the whole current of the appliance. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

4. Assertion (A): A sky-diver falls at constant speed after opening the parachute. Reason (R): The forces on her are balanced. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

5. Assertion (A): Dams are made thicker at the bottom. Reason (R): Water pressure increases with depth. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

6. Assertion (A): A gas can be compressed easily but a solid cannot. Reason (R): Gas particles are widely separated while solid particles are tightly packed. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

7. Assertion (A): Brass is a mixture of copper and zinc. Reason (R): The proportions of copper and zinc in brass can vary. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

8. Assertion (A): A soft drink goes flat when left open. Reason (R): The solubility of a gas in a liquid decreases as pressure decreases. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

9. Assertion (A): A concave mirror is used as a shaving mirror. Reason (R): It forms an enlarged erect image of the face when held close. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. Assertion (A): A leap year has 366 days. Reason (R): The Earth takes about 365.25 days to complete one revolution around the Sun. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

11. Assertion (A): Energy flow in an ecosystem is unidirectional. Reason (R): Energy lost as heat cannot be reused by producers. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

12. Assertion (A): The temperature in the stratosphere rises with height. Reason (R): The ozone layer absorbs ultraviolet rays from the Sun. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

13. Seeds in cups: A (dry cotton), B (wet cotton, kept in light), C (wet cotton, kept in dark). Which pair isolates the effect of light? 1

(a) A and C **(b)** B and C
(c) A, B and C together **(d)** A and B

14. A student observes an onion peel with 10x eyepiece and 10x objective, and the cell length in the picture is 5 mm. The real length is: 1

(a) 0.05 mm **(b)** 0.5 mm **(c)** 0.005 mm **(d)** 5 mm

15. A town has 20,000 people. If 15,000 are vaccinated, the percentage vaccinated is: 1

(a) 25% **(b)** 50% **(c)** 15% **(d)** 75%

16. Four cells in a circuit are connected one way and the bulb glows. If one cell is reversed, the bulb: 1

(a) glows brighter **(b)** melts **(c)** glows dimmer **(d)** does not change

17. **Assertion (A):** A solution of salt in water does not scatter a beam of light. 1
Reason (R): The particles in a true solution are extremely small.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

18. **Assertion (A):** The image formed by a plane mirror cannot be obtained on a screen. 1
Reason (R): The image in a plane mirror is virtual.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

19. **Assertion (A):** The shadow of a vertical stick is shortest around noon. 1
Reason (R): At noon the Sun is highest in the sky.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

20. **Assertion (A):** Green plants are called producers. 1
Reason (R): Plants make their own food by photosynthesis using sunlight.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 6 × 2 = 12

Answer in about 20–30 words each.

21. The core of the Earth accounts for 15 per cent of the Earth's volume. If the total volume is about 1.08 x 10^12 cubic km find the volume of the core (approx). 2

22. Give two differences between accuracy and precision. 2

23. Why is Amoeba able to change its shape but a plant cell cannot? 2

24. Why do chronic diseases like diabetes require lifelong care? 2

25. Why do two bulbs in series glow dimly compared with a single bulb? 2

26. Why does a block slide down a slope faster when the slope is polished? 2

SECTION — C (Short Answer Type) — 6 × 3 = 18

Answer in about 40–60 words each.

27. A bulb is connected through a very long thin wire to a cell. The bulb is dim. Explain and suggest improvement. 3

28. Explain why the friction between a book and the table does not depend much on the area of its face for the same weight. 3

29. Explain why the speed of wind is greatest near the eye wall. 3

30. Compare ice, water and steam in a table in terms of energy, spacing and motion of particles. 3

31. Explain why diamonds do not conduct electricity but graphite does. 3

OR

Explain how crystallisation purifies copper sulphate. 3

32. A student uses a concave mirror as a make-up mirror. She sees an enlarged, erect face when it is close but an inverted face when she moves far away. Explain. 3

OR

Why do countries have different time zones? How many hours of difference per 15 degrees of longitude? 3

SECTION — D (Long Answer Type) — 6 × 5 = 30

Answer in about 100–150 words each.

33. **Case study.** Harpreet pulls a box of mass 20 kg across a floor with a horizontal force of 60 N. The force of friction between the box and the floor is 40 N. When he puts rollers under the box, the friction drops to 10 N. 5

(i) Find the net force on the box in the first case.
(ii) Is the box moving or at rest?
(iii) State the direction of friction.
(iv) Find the net force with rollers and explain why rollers help.

34. **Case study.** Mohan weighs 600 N. When he stands on both feet, each foot has an area of 0.01 square metre in contact with the ground. When he lifts one foot, he stands on one foot only. 5

(i) State the formula for pressure.

(ii) Find the pressure when he stands on both feet.

(iii) Find the pressure when he stands on one foot.

(iv) A camel's wide feet do not sink in the desert sand, but a person's narrow heels do. Explain.

35. Explain the particle picture of an element, compound and mixture with the example of oxygen, water and air, and the numbers of atoms. 5

36. Explain how purified water is obtained from a salt solution by distillation, including the apparatus and principle, and why distilled water is used in labs. 5

OR

Describe in detail the working of the eye as an optical instrument and explain defects of vision, with remedies and lens power ideas. 5

37. A traveller sees the Pole Star at 30 degrees altitude. State his latitude, say where the Pole Star would appear at the equator and at the North Pole, and what happens to its height as he travels north. 5

OR

Discuss how rising CO2 in the atmosphere affects plants, oceans, ice and weather, and what individuals can do. 5

38. Compare the effects of increased CO2 on the oceans (acidification), the atmosphere (warming) and plants (growth). Discuss the benefits and the harms. 5

OR

A student's results for plant height (cm) after 4 weeks with watering of 0, 20, 40, 60, 80 mL were 3, 8, 12, 13, 9. Describe the pattern, suggest an explanation and a next experiment. 5

END OF QUESTION PAPER **Total: 80 Marks**

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